CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2023 Examination | **Paper Code:** 18023 | **Complete 100% Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION – A (VERY SHORT ANSWER QUESTIONS • 3 MARKS EACH • ALL 5 COMPULSORY)

Q1. Describe the various components of Data Communication. [3 Marks]

Answer:

Data communication is the exchange of data between two devices via a transmission medium. It comprises five essential components:

Component	Description & Role
1. Message	The information or payload to be communicated (Text, Numbers, Pictures, Audio, Video).
2. Sender (Transmitter)	The source device originating and transmitting the data message (PC, Workstation, Smartphone).
3. Receiver	The destination device that receives the transmitted data (Server, Printer, Mobile terminal).
4. Transmission Medium	The physical communication conduit carrying the signal from sender to receiver: Guided (Twisted pair, Coaxial cable, Fiber optic) or Unguided (Radio waves, Microwaves, Infrared).
5. Protocol	A formal set of governing rules that coordinate data transmission syntax, semantics, and timing (e.g., TCP/IP, HTTP).

Q2. Define Bandwidth and Network Congestion. [3 Marks]

Answer:

1. Bandwidth:
The maximum data transfer capacity of a communication channel over a given interval. In digital systems, it is measured in **bits per second (bps, Mbps, Gbps)**; in analog systems, it represents the frequency range (in Hertz, Hz) between upper and lower cutoff frequencies ($\texttt{Bandwidth} = f_{high} - f_{low}$).

2. Network Congestion:
A state where network nodes (routers/switches) carry more data packets than they can accommodate. When offered traffic load exceeds channel capacity, router queues overflow, causing **packet queuing delay, packet loss, and throughput collapse**.

Q3. Distinguish between MAC Address and IP Address. [3 Marks]

Answer:

Parameter	MAC Address (Physical Address)	IP Address (Logical Address)
OSI Layer	Data Link Layer (Layer 2).	Network Layer (Layer 3).
Address Length & Format	**48 bits (6 bytes)**; written in 12 hexadecimal digits (e.g., `00:1A:2B:3C:4D:5E`).	**32 bits** in IPv4 (e.g., `192.168.1.1`); **128 bits** in IPv6 (hexadecimal).
Permanence & Assignment	Permanently hard-coded into Network Interface Card (NIC) by manufacturer (OUI).	Dynamically (DHCP) or statically assigned based on network location.
Scope of Utility	Local node-to-node frame delivery within the same LAN segment.	Global end-to-end packet routing across multiple networks/Internet.

Q4. Differentiate between a Hub and a Switch. [3 Marks]

Answer:

Feature	Network Hub	Network Switch
Operating OSI Layer	Physical Layer (Layer 1).	Data Link Layer (Layer 2) / Multilayer (Layer 3).
Data Transmission Mode	**Broadcasts** all incoming frames to every connected port indiscriminately.	**Unicasts** frames selectively to the specific destination port using MAC table.
Collision Domains	All ports share a single collision domain (high collision risk).	Each individual port forms its own dedicated collision domain.
Bandwidth Sharing	Total bandwidth is shared among all connected active nodes.	Dedicated, full-rate bandwidth provided to each port (full-duplex).

Q5. What is Subnetting? Give an example with CIDR calculation. [3 Marks]

Answer:

Subnetting is the practice of logically partitioning a single large IP network into smaller, distinct subnetworks (subnets) by borrowing host bits for the network ID. This reduces broadcast domains, improves routing efficiency, and conserves IP addresses.

Example: Subnetting 192.168.1.0/24 into 4 Subnets

- To create 4 subnets: Borrow 2 bits from host portion ($2^2 = 4$ subnets).
- New prefix length: `/24 + 2 = /26` ⇒ Subnet Mask: `255.255.255.192`.
- Block Size: `256 - 192 = 64` addresses per subnet (62 usable hosts).

Subnet #	Network ID	Usable Host IP Range	Directed Broadcast IP
Subnet 1	`192.168.1.0`	`192.168.1.1 - 192.168.1.62`	`192.168.1.63`
Subnet 2	`192.168.1.64`	`192.168.1.65 - 192.168.1.126`	`192.168.1.127`
Subnet 3	`192.168.1.128`	`192.168.1.129 - 192.168.1.190`	`192.168.1.191`
Subnet 4	`192.168.1.192`	`192.168.1.193 - 192.168.1.254`	`192.168.1.255`

SECTION – B (SHORT ANSWER QUESTIONS • 7.5 MARKS EACH • ATTEMPT ANY TWO)

Q6. Describe the role of the Transport Layer in the OSI Model. [7.5 Marks]

Answer:

The **Transport Layer (Layer 4)** serves as the boundary between hardware-oriented subnet layers and software application layers, providing true **end-to-end, process-to-process reliable data delivery**.

Core Function	Detailed Mechanism & Protocol Realization
1. Service-Point (Port) Addressing	Differentiates concurrent application processes executing on the same host using 16-bit Port numbers (e.g., HTTP: 80, HTTPS: 443, DNS: 53, SSH: 22).
2. Segmentation & Reassembly	Splits massive application messages into manageable chunks called **Segments**, appends sequence numbers, and reassembles them at receiver in exact order.
3. Connection Management	Manages connection-oriented transmission via 3-way handshake (TCP) or fast connectionless datagram delivery (UDP).
4. Flow Control	Uses **Sliding Window Protocol** to prevent a fast transmitter from overwhelming a slow receiver's buffer space.
5. Error Control	Enforces end-to-end integrity using 16-bit **Checksums** and Automatic Repeat reQuest (ARQ) retransmissions for dropped/corrupted segments.

Q7. Explain Network Topologies (Star, Mesh, Bus, Ring) with diagrams and comparisons. [7.5 Marks]

Answer:

Network Topology defines the geometric and logical arrangement of communication links and nodes in a network.

1. Star Topology:

All nodes connect via dedicated point-to-point links directly to a central multiport device (Hub or Switch).

Layout: `[Node A, Node B, Node C, Node D] → [Central Switch / Hub]`

Pros: Easy to install, failure of one node does not impact others. **Cons:** Central hub is a single point of failure.

2. Mesh Topology (Full Mesh):

Every node has a dedicated point-to-point link to every other node in the network.

Link Formula: For `n` nodes, requires $\frac{n(n-1)}{2}$ physical links and `(n - 1)` I/O ports per device.

Pros: Maximum redundancy, zero traffic contention, highly fault-tolerant. **Cons:** Extremely expensive cabling and port requirements.

3. Bus Topology:

All nodes share a single common linear backbone cable (coaxial cable with terminators at both ends).

Layout: `Terminator ── [Node 1] ── [Node 2] ── [Node 3] ── Terminator`

Pros: Minimal cabling needed. **Cons:** A single break in backbone takes down entire network; hard to troubleshoot.

4. Ring Topology:

Each device connects directly to its two immediate neighbors, forming a closed continuous loop.

Layout: `Node 1 → Node 2 → Node 3 → Node 4 → Node 1 (Token Passing)`

Pros: Orderly token transmission, no collisions. **Cons:** Break in ring disrupts network (unless dual counter-rotating ring like FDDI is used).

Q8. Explain Congestion Control Algorithms (Leaky Bucket and Token Bucket). [7.5 Marks]

Answer:

Traffic shaping algorithms regulate the bursty flow of packets injected into the network to mitigate congestion:

Parameter	Leaky Bucket Algorithm	Token Bucket Algorithm
Core Operating Principle	Bursty packet arrivals enter a finite buffer; packets leak out at a **strictly constant, rigid rate**.	Tokens generate into bucket at regular rate `r` up to capacity `C`. Packets transmit by consuming tokens.
Bursty Traffic Handling	**Smooths out all bursts completely**; discards packets if bucket capacity overflows.	**Allows controlled bursts** of packets up to the accumulated token capacity `C`.
Output Transmission Rate	Strictly uniform, deterministic constant rate regardless of burstiness.	Variable rate bounded by average rate `r` and peak burst capacity.
Packet Discard Policy	Discards incoming packets when buffer memory fills up.	Discards tokens when token bucket fills; packets queue or wait for fresh tokens.

SECTION – C (LONG ANSWER QUESTIONS • 15 MARKS EACH • ATTEMPT ANY THREE)

Q9. Explain the TCP Three-Way Handshake with a sequence diagram. [15 Marks]

Answer:

The **TCP 3-Way Handshake** is the synchronization mechanism used by Transmission Control Protocol to establish a reliable, full-duplex virtual connection between client and server before data exchange begins.

TCP 3-Way Handshake Connection Sequence:

Step	Direction	Control Flags & Sequence #	Action & Purpose
Step 1 (SYN)	Client → Server	`SYN = 1` `Seq = X` (Client ISN)	Client requests connection and announces its initial sequence number X. Client enters **SYN-SENT** state.

Step 2 (SYN-ACK)	Server → Client	`SYN = 1, ACK = 1`
		`Seq = Y` (Server ISN)
		`Ack_Num = X + 1`
			Server acknowledges client's request (`Ack = X + 1`) and provides its own sequence number Y. Server enters **SYN-RCVD**.

Step 3 (ACK)	Client → Server	`ACK = 1`
		`Seq = X + 1`
		`Ack_Num = Y + 1`
			Client acknowledges server's sequence number (`Ack = Y + 1`). Connection enters **ESTABLISHED** state on both endpoints.

Significance: Guarantees both parties agree on initial sequence numbers, confirms bidirectional communication capability, and prevents old duplicate connection requests from causing false sessions.

Q10. Draw and explain the 7 Layers of the ISO/OSI Reference Model. [15 Marks]

Answer:

The **Open Systems Interconnection (OSI)** reference model is a conceptual 7-layer architectural framework developed by ISO for standardized computer communication:

Layer	Layer Name	PDU	Primary Functions & Responsibilities	Standard Protocols
Layer 7	**Application**	Data	Provides user interface and network service abstraction for software apps.	HTTP, HTTPS, FTP, SMTP, DNS, DHCP
Layer 6	**Presentation**	Data	Data syntax translation, compression, encryption and decryption (SSL/TLS).	JPEG, MPEG, ASCII, TLS/SSL
Layer 5	**Session**	Data	Dialog control, token management, session checkpointing and synchronization.	RPC, NetBIOS, PPTP
Layer 4	**Transport**	Segment	End-to-end process delivery, port addressing, flow control (sliding window), error recovery.	TCP, UDP
Layer 3	**Network**	Packet	Logical IP addressing, subnet routing, packet forwarding across networks.	IPv4, IPv6, ICMP, OSPF, BGP
Layer 2	**Data Link**	Frame	Node-to-node frame delivery, physical MAC addressing, CRC error detection, CSMA/CD.	Ethernet (802.3), Wi-Fi (802.11), PPP
Layer 1	**Physical**	Bit	Transmission of raw bitstream over physical media, signal encoding, voltages.	RS-232, RJ-45, V.35, Cables

Q11. Explain Shannon Capacity Theorem with a solved numerical problem. [15 Marks]

Answer:

Shannon's Capacity Theorem establishes the theoretical upper limit on the maximum data transmission rate (Channel Capacity C) across a noisy communication channel with white Gaussian thermal noise.

> Shannon Formula: $C = B \bullet \log_2 \&left;(1 + \frac{S}{N} \&right;)$ (in bits per second, bps)
>
> Where:
> - `C` = Maximum theoretical channel capacity in bits per second (bps)
> - `B` = Channel bandwidth in Hertz (Hz)
> - `S / N` = Signal-to-Noise Ratio (SNR as a linear power ratio)

Solved Numerical Problem:

> **Problem Statement:** A noisy telephone channel has a bandwidth of *B = 3000 Hz* and a Signal-to-Noise Ratio of SNR_{dB} *= 35 dB*. Calculate the theoretical maximum channel capacity.
>
> **Step 1: Convert SNR from decibels (dB) to linear ratio:**
>
> $$\text{SNR}_{\text{dB}} = 10 \bullet \log_{10}(S/N) = 35 \Rightarrow \log_{10}(S/N) = 3.5 \Rightarrow \frac{S}{N} = 10^{3.5} \approx 3162.28$$
>
> **Step 2: Apply Shannon's Formula:**
>
> $$C = 3000 \bullet \log_2(1 + 3162.28) = 3000 \bullet \log_2(3163.28)$$
>
> Using change of base: $\log_2(3163.28) = \frac{\log_{10}(3163.28)}{\log_{10}(2)} = \frac{3.50014}{0.30103} \approx 11.6272$
>
> $$C = 3000 \times 11.6272 \approx 34{,}881.6 \text{ bps} \approx 34.88 \text{ kbps}$$

Theoretical Channel Capacity: C ≈ 34.88 kbps

Q12. Explain CSMA/CD in Ethernet and compare with CSMA/CA in Wi-Fi. [15 Marks]

Answer:

Carrier Sense Multiple Access (CSMA) protocols require nodes to "listen before talking" on a shared broadcast transmission medium.

Comparison Feature	CSMA/CD (Collision Detection)	CSMA/CA (Collision Avoidance)
Standard & Medium	IEEE 802.3 (Wired Ethernet cables).	IEEE 802.11 (Wireless Wi-Fi radio channels).
Collision Strategy	Transmits while listening; if collision signal is detected, stops immediately.	Avoids collisions prior to transmission via reservation handshakes and delays.
Collision Response	Transmits a **Jam Signal** to notify all stations, then applies **Binary Exponential Backoff**.	Cannot detect collision while sending (half-duplex RF); infers collision if no ACK received.
Control Mechanism	Collision detection hardware listening for elevated cable voltage.	**RTS (Request to Send) / CTS (Clear to Send)** handshakes + Inter-Frame Spacing (DIFS/SIFS).

Q13. Compare Circuit Switching, Packet Switching, and Message Switching. [15 Marks]

Answer:

Switching is the methodology by which nodes communicate across multiple intermediate network nodes without direct point-to-point physical links.

Feature	Circuit Switching	Packet Switching	Message Switching
Dedicated Physical Path	**Yes**; dedicated end-to-end copper/fiber circuit reserved for session duration.	**No**; dynamic independent packet routing over shared virtual links.	**No**; whole message hops from node to node across shared lines.
Call Setup Phase	Required prior to any data transmission. Adds initial setup latency.	Zero connection setup latency (in Connectionless Datagram mode).	No setup phase needed.
Bandwidth Utilization	Low/wasteful; reserved circuit capacity is idle during conversational silence.	**Very High**; dynamic statistical multiplexing shares channel bandwidth.	Moderate; links occupied only while whole message transmits.
Store & Forward Mechanism	None; continuous real-time bitstream.	Packets buffered at each router and forwarded. Minimal latency.	Entire message buffered on intermediate disks; very high delay.
Primary Application	Traditional Landline Telephony (PSTN), cellular voice calls.	**The Modern Internet (IP)**, VoIP, streaming, web browsing.	Telegraph, Telex, historic military communication systems.